

Inkscriptions

Permanent Artifacts Inked on Chain



Inkscriptions is a decentralized application (dApp) built on the Ink L2 blockchain. It allows users to permanently etch text and ASCII art directly into the blockchain history and subsequently “wrap” these raw data artifacts into tradeable, visual NFT cards.

Core Features

1. Permanent On-Chain Etching

The core functionality of Inkscriptions is the ability to store data permanently. Unlike standard NFTs that often point to external storage such as IPFS, an Inkscription’s data is stored in the **calldata** of a transaction.

- **Mechanism:** Users burn 1000 BERT tokens to send a transaction. The text or ASCII art is appended to the transaction input data.
- **Result:** The message becomes an immutable part of the blockchain’s history, censorship-resistant and viewable forever by decoding the transaction hash.

2. Advanced Image-to-ASCII Converter

To facilitate on-chain art, the app includes a powerful, client-side generation tool:

- **Custom Rendering:** Converts uploaded images into high-fidelity ASCII art.
- **Fine-Tuned Controls:** Users can adjust **Width** (up to 125 chars), **Contrast**, **Brightness**, **Gamma**, and **Density** to perfect the output.
- **Character Sets:** Includes diverse rendering modes like *Extended High* (for detail) and *Retro Mixed* (combining standard text with Code Page 437 block characters) for a classic ANSI art aesthetic.

3. Gamified Rarity System

Every Inkscription is assigned a rarity tier based on the luck of the draw—specifically, the trailing characters of its unique Transaction Hash.

- **Common:** Standard hash.
- **Uncommon:** Ends in 2, 4, 6, or 8.
- **Rare:** Ends in ‘b’, ‘e’, or ‘7’.

- **Epic:** Ends in '0' or 'f'.
- **Legendary:** Ends in double characters (e.g., 'aa', '88').
- **Mythical:** Ends in triple characters (e.g., '777', 'bbb').

4. Oracle-Verified NFT Minting

While the raw data lives in the transaction, users can mint their Inkscription and choose a custom total supply to bring it into the NFT ecosystem.

- **The Trading Card:** The app dynamically generates a high-resolution Trading Card image using HTML5 Canvas. This card visualizes the ASCII art, the rarity tier (with color-coded borders and glows), and the mint details.
- **Procedural Art:** *Mythical* tier cards generate a unique "Ink Nebula" background effect every time they are rendered.
- **Oracle Bridge:** A secure Oracle validates the original transaction hash and metadata before allowing the user to mint the official ERC-1155 NFT.

5. Immersive UI/UX

The interface is designed with a modern "Web3 Pop" aesthetic:

- **Glassmorphism:** Cards use a frosted glass effect (backdrop-filter) allowing the dynamic background to bleed through.
- **Dynamic Visualization:** The gallery and fullscreen views adapt to the content's rarity. If an item is **Legendary**, the ASCII text itself renders in Gold; if **Mythical**, it renders in Platinum.
- **Typography:** Utilizes a hierarchy of fonts (Bungee for graffiti-style titles, Fredoka for clean readability) to balance fun and function.

Technical Stack

- **Frontend:** Vanilla HTML5, JavaScript (ES6+), and Tailwind CSS.
- **Blockchain Interaction:** Ethers.js (v6) for wallet connection, gas estimation, and contract interaction.
- **Networks:** Optimized for **Ink L2 Mainnet** (Low gas fees allow for data-heavy transactions).
- **Smart Contracts:**
 - **BERT Token:** Standard ERC-20 (Burn mechanism).
 - **Inkscriptions NFT:** ERC-1155 (Minting and Metadata management).
- **Backend Services:** Firebase (for caching/indexing the gallery) and a Cloud Function Oracle (for mint verification).

User Flow

1. **Connect:** User connects their Web3 wallet (Rabby, MetaMask) to Ink L2.

2. **Compose:** User types a message or uses the "Image to ASCII" tool to generate art.
3. **Etch:** User approves the transaction, burning 1000 BERT. The data is inscribed into the transaction's input data.
4. **Discover:** The new Inkscription appears in the Live Gallery. Its rarity is revealed based on the Transaction Hash generated. Users can see a preview of the NFT.
5. **Mint (Optional):** The owner can choose to mint the artifact as an NFT, choosing a supply between 1 and 1,000,000. The app generates the visual card, the Oracle verifies the data, and the user receives a tradeable ERC-1155 token.